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Description of change: 1.Changed title from all caps to sentence case. 2.Added navigation markers throughout document. 3.Aligned PRI/PSI terminology throughout, including in title of document. 4."Serious Injury / Serious Deterioration of Health" changed to "Serious Injury / Serious Deterioration of the Status of Health". 5.New clarifying questions added to "Death" category. 6.Additional guidance sections updated in "Serious Events", "Erroneous Results", and examples updated in text section after "Erroneous Results" tables. 7.Added note for self-administered treatment in "Serious Events" table. 8.Updated Notes before guidance table for Erroneous Results. 9.Added "Hook" in Clinical Chemistry that invalidates results table. 10.Calculation of sodium values example moved from Erroneous results table to Guidance on Calculation of Erroneous Results section. 11.Typographical and grammar changes throughout document. (REQ01311R)

Justification for change: 1.Alignment with other documents' style. 2.Enhancement for ease of use. 3.Alignment across and within documents. 4.Clarification of intent. 5.Enhancement. 6.Clarification and alignment, to reduce incorrect / unnecessary escalation of PRIs. 7.Alignment with UK legislation requirements for blood glucose testing. 8.To align with changes made in MQMS-PM-GSP-05-CC-02 table headers. 9.Additional parameter. 10.Clarification. 11.Improvements. (REQ01311R)

Electronic Signatures:

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Purpose

This document provides additional information and examples to assist with determining if a complaint requires a classification as a potentially reportable incident (PRI) and potential safety impact (PSI). This document does not cover every possible scenario that may be encountered in a complaint. If there are questions or concerns regarding a specific complaint and PRI or PRI/PSI classification, please consult with your Local Safety Officer (LSO) or appropriate subject matter expert (SME).

NOTE: The Food and Drug Administration (FDA) can request additional requirements in an Emergency Use Authorization (EUA), which may require cases to be escalated as PRI.

Navigation

Please click on the links below to access the guidance for each of the PRI/PSI categories:

1. [Death](#)
2. [Serious Injury / Serious Deterioration of the Status of Health](#)
3. [Erroneous Results – General Section](#)
 - a. [Information Regarding Valid Sample Comparisons for Erroneous Results](#)
 - b. [Handling Flagged and Unbelievable Results](#)
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4. [Malfunction – General Section](#)

- a. [Near Patient Care \(NPC\) Specific Malfunctions](#)
 - b. [Product-Specific Malfunctions Requiring Escalation as PRI](#)
- 5. [Cybersecurity or Data Breach](#)
 - 6. [Notification of the Competent Authority](#)

NOTE: For complaints where the affected product is a Research Use Only (RUO), Laboratory Use Only (LUO), Analyte Specific Reagents, Industrial Business, or any additional non-In Vitro Diagnostic test commercially distributed by Roche Diagnostics, escalate complaint as a PRI/PSI only if the issue led to death, serious injury, and/or serious harm; all other PRI/PSI categories do not apply for these tests.

Death

Customers may report that a patient(s) has died. When “death”, “deceased”, “perished”, or “passed away” is mentioned by the customer, request the necessary information to determine if death may have been related to a Roche device. Information to request may include, but is not limited to:

- Clarify how the Roche product may have been involved in the death.
- What information suggests results from the device were incorrect?
 - Was testing repeated and what were the results, including non-Roche methods?
 - What was/were the date(s) of testing?
- What actions/decisions were taken or not taken that are alleged to have caused or contributed to the death?
- If a Roche device could not be used and it is alleged this caused or contributed to the death of a patient, was another device or lab available to perform testing?
- What other information suggests the death was due to a Roche product?
 - Medical information, professional allegation, additional information.
 - Where did the event occur (Intensive Care Unit [ICU], emergency room [ER], ambulance, home, other)?
 - Medical history of the patient prior to the reported death.
 - Was the death expected (e.g., palliative care or hospice)?
 - What was the specific cause of death (according to the Death Certificate)?

In some situations, the customer may allege explicitly that the device had something to do with the death, which must be escalated to PSI. For example, incorrect results led to incorrect treatment or diagnosis that was alleged to have led to or caused the death of a patient.

In some cases, the death of a patient may be mentioned, however, there may be no allegation that a Roche device caused or contributed to the death. In these cases, where the death is clearly not related to the Roche device there is no requirement for escalation to PSI. For example, death may have been caused by the inevitable course of a disease or acute event. The potential failure of laboratory results or the device may have been only a coincidence with the death.

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Serious Injury/Serious Deterioration of the Status of Health

When serious injury or serious deterioration of health is mentioned by the customer, request the necessary information to determine if harm may have been related to a Roche device. Information to request may include, but is not limited to:

- What actions or decisions were taken (e.g., incorrect medication changes, treatment incorrectly initiated or stopped) that are alleged to have caused or contributed to harm to the patient?
- What actions or decisions were not taken (e.g., no treatment initiated, necessary medications not provided) that are alleged to have caused or contributed to harm to the patient?
- Historical events of actions, such as dates and times (hours, days, etc.) of testing, decision, and treatment.
- What was the confirmed diagnosis of the patient?
- What type of testing was performed to confirm the diagnosis?
- What other information suggests that a Roche product may have been a contributing factor in the serious injury/serious deterioration in state of health?
 - Medical information, professional allegation, other information.
 - Where did the event occur (ICU, ER, Ambulance, Home, other)?
 - Medical history of the patient, including medication(s), prior to the reported serious injury or serious deterioration of health.
 - Current status of the patient on the day of the event (e.g., stable, critical, improving).
- Was the device used according to Roche specifications?

The following table describes examples of situations that may require classification as PRI and PSI, along with additional guidance to support the decision-making process.

NOTE: *A serious event requires the patient or operator to receive professional medical treatment, and/or have significant harm or injury. If there is no confirmation of harm to/impact on the patient, do not escalate as PSI. However, depending on the incident, a classification of PRI may still be required.

Types of Serious Events	Examples	Additional Guidance
Life-threatening illness, injury, or harm.	• Incorrect diagnosis causing a life-threatening situation. • Stroke, transient ischemic attack (TIA), intracranial / subdural / subarachnoid hemorrhage. • Myocardial infarction, cardiac arrest, cardiopulmonary resuscitation, cardiac event. • Hemorrhage, blood clot, embolism, pulmonary embolism, thromboembolism, thrombosis, thrombus. • Delay in diagnosis of cancer. • Reduction in life expectancy. • Delay in test result that is needed during surgery. (The delay during the surgery is considered the harm in this situation.) • Delay in treatment causes harm to the patient. • Organ transplant that cannot be performed or is delayed due to an erroneous or delayed result. • Delay in patient diagnosis leads to harm or worsening of the patient's condition.	• Delays in results that do not harm the patient, do not require escalation as PSI, except for a delay in results during surgery, as this delay inherently constitutes harm (e.g., [1] due to prolonged anesthesia or severe health consequences due to misled medical operational decision; [2] the surgeon decides to proceed without results, e.g., parathormone). • Delay in elective treatment, such as fertility treatment, does not require escalation as PSI. • When a device is unavailable for use, a PSI is not required unless there is harm to the patient. Information to collect: • How long was the delay (minutes, hours, days, a week, or more)? • How did the delay affect the patient and was there harm to the patient? • Was another device available to perform testing?

Types of Serious Events	Examples	Additional Guidance
Permanent impairment to a body function or structure.	• Patient has a stroke with paralysis. • Receipt of blood donation with positive viral serology. • Irreversible eye damage due to exposure to Roche product. • Operator lost finger while operating Roche device. • Heart insufficiency/failure after heart attack (myocardial infarction). • Diagnosis and appropriate treatment for viral infectious disease missed due to erroneously negative/non-reactive Ab, Ag, or DUO/COMBI/result. • Misleading diagnostic and therapeutic decision on cancer disease. • Congenital, physical or mental impairment or birth defect.	Scars are not considered permanent damage that requires escalation as serious injury (PRI/PSI), except when they cause loss of sensitivity or restrictions in motility.
Surgical intervention.	• One or more surgical procedures were required, or an existing procedure was changed due to incorrect device performance. • Incorrect increase in inflammation markers, leading to the incorrect assumption of abdominal infection after surgery. • Incorrect increase in the course of tumor markers, leading to surgical exploration, endoscopy, etc.	N/A
Invasive testing procedures based upon erroneous results.	• Exploratory surgery due to incorrect result. • New or repeat biopsy due to incorrect result. • Left heart catheter (arterial puncture, high dosage x-ray, iodine contrast). • Unnecessary amniocentesis.	The following <u>do not</u> require escalation as PRI/PSI: • Extra blood draws • Ultrasounds • Vaginal ultrasounds, except in cases of preterm contractions and/or preterm rupture of amniotic sac • Intravenous line placement

Types of Serious Events	Examples	Additional Guidance
Illness or injury resulting from incorrect treatment provided, treatment incorrectly withheld, or treatment delayed or changed.	• Incorrect treatment or dosage of medication (including missing treatment, under dose, or overdose) suggested, provided, or given which causes harm. • Incorrect diagnosis which causes harm to the patient. • Patient has a seizure, coma, faints or cardiac events due to incorrect treatment. • Inadequate/inappropriate/unnecessary radiation dosage during procedures. • Fetal harm due to missed diagnosis of infection, congenital diseases, missed diagnosis of maternal infection. • Harm of newborn due to missed diagnosis of infection or incorrect assessment of infectious status of mother before delivery (missed cesarean section). • Receiving incorrect or incompatible whole blood or a blood component. • Nosebleed requiring blood transfusion.	Bleeding events with a clear cause unrelated to a Roche device are not PSI. Information to collect: • What incorrect treatment was received due to the device? What treatment was given before and after the event? • How did the incorrect treatment affect the patient and what was the harm to the patient? • How did the inappropriate treatment affect the patient and what was the harm to the patient?

Types of Serious Events	Examples	Additional Guidance
Unexpected medical intervention due to a Roche device.	• Patient required an infusion of whole blood or a blood component directly into the bloodstream. • Patient required additional medication or an additional dose of existing medication. • Incorrect results from the Roche device led to administration of Vitamin K, Lovenox, Heparin, blood transfusion or fresh frozen plasma for coagulation issues.* • Incorrect results from the Roche device led to administration of Insulin or glucose for diabetic management.* <i>*Note: Also applies to self-administered treatment.</i> • Performing cardiopulmonary resuscitation (CPR). • Prophylaxis treatment of infectious diseases (blood-borne pathogens) due to incorrect result from Roche device.	When the meter results and other testing indicate the treatment given was the correct measure with no harm to the patient, escalation as PSI is not required.

Types of Serious Events	Examples	Additional Guidance
Direct injuries to the patient or operator of the Roche device that require the patient or operator to receive professional medical attention.	- Cuts that require stitches. - Wounds that require treatment for infection. - Bruises, burns including chemical burns, or muscle injuries that require treatment by a medical professional. - Electrical shock that results in permanent impairment or treatment required to prevent permanent impairment. - Accidental needle sticks (e.g., sample or reagent probe, used lancet) where the individual receives treatment such as tetanus shots, other immunizations, prophylaxis or stitches. - User of device is splashed or sprayed with a reagent or other Roche product and the person receives medical treatment from a medical professional. - Broken glass or tubes containing patient samples inside of a Roche device cut a user (e.g., broken tube in centrifuge cuts the device user) requiring medical intervention. - Broken bone, ruptured tendon, torn ligaments, torn cartilage.	Scenarios <u>not</u> requiring escalation as PSI: - Treatment such as Band-Aids, Steri-strips (plasters), and minor first aid (washing of injured area). - Electrostatic discharge, with or without discomfort with no permanent impairment or treatment required to prevent permanent impairment. - Injuries requiring only infectious disease testing without prophylaxis treatment. - Where it is not known if professional medical treatment was received. - Cuts from cardboard or other packaging (these are not medical devices). - A minor sprain or tendonitis not requiring medical intervention other than basic first aid. - Splashes (non-infectious fluid, non-toxic, no acids) to user if only simple rinsing of the skin or eyes is required as this is not considered medical treatment. Scenarios that <u>may</u> require escalation as PSI: - Accidental needle sticks may also include injuries from other Roche devices or consumable components. These are PSI IF the patient receives medical attention as described. - Broken glass or tubes contained within a Roche device do not require escalation as PSI unless there is harm to the user.

Types of Serious Events	Examples	Additional Guidance
Allergic or toxic reactions due to contact with Roche product(s) (e.g., fumes, vapors, direct contact).	- A rash or skin irritation occurs due to contact with Roche product and treatment by a medical professional is required. - Headaches and nausea due to contact with Roche products and medical treatment by a medical professional is required. - Eye irritation due to contact with a Roche product and treatment by a medical professional is required.	- If a customer is splashed anywhere on their body or face with a material and washes the material off with no medical treatment, this does not require escalation as PSI. - Headaches and nausea that do not require treatment by a medical professional, do not require escalation as PSI.
Hospitalization due to health damage from the device.	- Patient is admitted to the hospital due to incorrect results or malfunction of the device. - Patient requires admission to intensive care unit or extension of stay in an intensive care unit. - Emergency room visit resulting in admission to the hospital.	- If a patient is hospitalized or their hospital stay is extended and there is no malfunction of a Roche device, this is not a PSI. - If a patient is seen in an emergency room and released without admission to the hospital, this is not a PSI. If treated and released, please refer to other examples within this section to ensure PSI is not needed.
Rehabilitation	Patient requires rehabilitative treatment to facilitate the process of recovery from injury, illness, or disease.	N/A

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Erroneous results

When erroneous results are mentioned by the customer, request the necessary information to determine if the results meet PRI criteria according to [MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results](#). Information to request may include, but is not limited to:

- Was repeat testing performed on the same sample or a different sample?
- Which result was considered correct?
- How much time passed before testing was repeated?
- Was the sample diluted and why was a dilution performed?

- Were there any flags or alarms with the results or from the instrument?
- Were erroneous results reported to the patient?
- Were erroneous results released to the medical record and/or medical personnel treating the patient?
- Was the erroneous result auto-validated and released, or manually checked before being issued?
- Was pre-analytical sampling handled per instructions for use (e.g., collection, proper tube, proper storage, proper centrifugation)?
- Were instructions for use followed for sample preparation and for testing?
- If in comparison to another method, what are the reference ranges (normal ranges) for the other method?

NOTE: Medical Record is defined as follows: The patient medical file to which the result of the test will be made available for viewing by those medical professionals treating or consulting on the patient outside of the laboratory performing the test. Multiple formats are possible, including, but not limited to: paper copy, electronic copy (e.g., Electronic Medical Record (EMR), Electronic Medical Form (EMF), verbal report, etc).

The Table below describes Erroneous Result scenarios, including worst-case scenarios and when a sample is tested multiple times, with additional guidance to support the decision making process.

NOTE: Do not use the scenarios and guidance below if there is an X in the 'No escalation for PRI ER Required' column in [MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results](#).

NOTE: If the results lead to Death, Serious Injury and/or Serious Harm, these cases must be marked as PRI/ PSI.

Scenario	Guidance
Customer gives estimated results but does not know exact results.	Evaluate for PRI using the estimated results.
Allegation of erroneous results and the customer does not provide results to Roche, test was not repeated, and/or no other information is provided that suggests the result was not correct.	Do not escalate as PRI and continue to process complaint per MQMS-PM-GSP-04 - Case Handling .
Allegation of erroneous result and the customer, end user, or physician states the results do not agree with the clinical status of the patient.	Escalate as PRI if allegation is supported by additional information, such as patient diagnosis, clinical status, clinical history, or other testing.
Allegation that results are “too low” or “too high” in comparison to a second result.	• If no results were given for the “too low” or “too high” values, do not escalate as PRI and continue to process complaint per MQMS-PM-GSP-04 - Case Handling. • If results were provided for evaluation, evaluate for PRI using the values provided.

Scenario	Guidance
A Roche Diagnostics pre-analytic device does not process samples properly, causing erroneous results to be produced.	The case should be escalated on the Roche Diagnostics pre-analytic device and marked PRI if the results meet the PRI criteria according to MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results .
A patient sample gives a negative Estrogen Receptor result and a positive Progesterone Receptor result and results were reported to a physician. The testing is repeated using another manufacturer's product and the result is negative. It is not known which result is correct.	- The criteria for Progesterone Receptor: False Positive or False Negative released to medical record. - The worst-case scenario would be a false positive Progesterone Receptor result released to the medical record. - This case would meet PRI criteria based upon the worst-case scenario because the Progesterone Receptor result could be a false positive reported to a physician.
A patient sample exhibits AFB-positive bacteria when tested with the BenchMark Special Stains AFB Staining Kit. However, the positive bacteria are in a different plane than the patient tissue and are located in areas not expected to be AFB positive (e.g., non-granulomatous areas), so the result is suspicious and is not reported to the patient's medical record. The test was repeated and found to be negative suggesting the initial positive result was likely due to a contaminant.	The case does not meet the PRI criteria because the incorrect positive Benchmark Special Stains AFB Staining Kit result was not believable and no erroneous results were released to the medical record.
A patient sample generates a result of TND for HIV-1 on CAP/CTM #1. On CAP/CTM #2, the same sample generates a titer of "10 ³ copies/mL" The expected result is unknown.	As it is not known if the TND result or the "10³ copies/mL" result is correct, the case should be marked as a PRI. Additional Guidance: - If the "10³ copies/mL" result was used as the "reported result" and the TND result was used as the "actual result" in the quantitative table for HIV-1, the case would not represent a PRI. - If the "TND" result was used as the "reported result" and the "10³ copies/mL" result was used as the "actual result", the case would meet PRI criteria.

Scenario	Guidance
Cross reactivity and interferences of samples which are not addressed in product labeling.	A patient sample is alleged to give incorrect results due to a medication the patient is on or a condition the patient has. NOTE: Cross reactivity and interferences of samples which are addressed in the product labeling are not required to be classified as PRI unless the results lead to Death, Serious Injury and/or Serious Harm. These cases must be marked as PRI/PSI.

The following Table describes **software issues** that may require escalation as PRI for erroneous results along with additional guidance to support the decision-making process:

Types of Erroneous Results Events	Examples	Additional Guidance
Sample mismatch.	A result is assigned to a wrong sample, wrong patient ID, wrong calibration, or other mismatches.	If the customer incorrectly entered/assigned the information and no patient was harmed, this is not a device malfunction and does not require escalation as PRI.
Test results are incorrectly modified and meet the PRI criteria according to MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results	Test results are multiplied by an incorrect conversion factor or formula.	If the customer incorrectly modified the test results and no patient was harmed, this is not a device malfunction and does not require escalation as PRI.
Wrong interpretation of the result due to a software malfunction.	- Interpretation of a PCR curve fails and reports positive instead of negative result. - Bolus Calculator dosing suggestion is incorrect due to incorrect value computation or conversion.	- If the mismatch meets the PRI criteria according to MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results, escalate as PRI. - Ask for potential medical consequences in treatment.
Test results incorrectly transferred by a Roche Diagnostics product to another software platform.	- Wrong transmission of test results to LIS / LIMS. - Change in glucose value when importing from a Roche glucose meter into mySugr app.	If the LIS/LIMS or other non-Roche middleware is known to be the cause of the issue, a PRI is not required.

Types of Erroneous Results Events	Examples	Additional Guidance
Incorrect measurement units that cause a difference in results when the difference using the correct measurement units meets screening list criteria.	Initial Result = mg/dL Repeat Result - mmol/L and mg/dL is incorrect When initial test results are converted from mg/dL to mmol/L, the difference in results in mmol/L meets the PRI criteria according to MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results .	When determining if results meet PRI criteria, ensure that both results are evaluated in the same unit of measure.
Instrument or software problems which cause in incorrect protocol/parameters to be used during testing	Wrong protocol used to generate results e.g., HCV protocol A is ordered but HBV protocol is run by the instrument and delivers HCV result.	• If operator chooses to use the wrong protocol, or incorrect sample preparation method, this would not be PRI. • If this is not a user error, it is a PRI.
Part of the result is truncated, leading to wrong values	• A result of 54 is displayed and/or printed as a 5. • Missing separator (comma, decimal point) displays or printed as 26 but value is 2.6.	N/A
Allegation of erroneous results in a standalone software product (software installed on a computer/ server/ control unit) which is provided either from Roche or from the customer (e.g., navify Lab Operations). The result is valid (even if erroneous) and obtained within a production environment. Note: Applications on cobas Pulse and navify® Mutation Profiler are excluded.	• A patient receives an incorrect result because the software product has incorrectly applied a formula result transformation. • Incoming results are incorrectly truncated, leading to an incorrect value. • Where the software is intended to enable correcting input errors or to expect incorrect results as part of its function, an incorrect result cannot be edited or corrected due to an issue, even if the incorrect result was ultimately derived from erroneous input.	Standalone software issues where erroneous results occur and work in an overarching way* require escalation as PRI, even if PRI criteria defined in MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results are not met. Products assessed to work in a non-overarching way: navify Algorithm Suite and related Algos. *Products work in an overarching way: issues may affect any tests used in the product.

Types of Erroneous Results Events	Examples	Additional Guidance
Allegation of erroneous results in navify[®] Mutation Profiler when reported to the physician or patient. Definition of erroneous result is the following: - AMP/ASCO Tier IA alterations and/or genomic signatures erroneously assigned to Tier 1B or lower tiers, in the presence of the approved therapy in a specific geographic region. - AMP/ASCO non-Tier IA alterations or genomic signatures erroneously assigned to Tier IA, in the absence of approvals of an appropriate therapy in a geographic specific region. - AMP/ASCO Tier IB alterations and genomic signatures erroneously assigned to Tier IIC or lower tiers. - AMP/ASCO below Tier IB alterations and genomic signatures erroneously assigned to tier IB. - Variant erroneously matched to a clinical trial that it should not have matched to. - Variant not displayed at all in the user interface (UI) and/or report when it should have been displayed and had a classification of IIC or higher or been matched to a clinical trial.	- A patient is prescribed a therapy that should not have been associated with a patient mutation. - The navify Mutation Profiler report does not display an actionable mutation (Tier IIC or higher) which was called by the secondary analysis pipeline.	

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Information Regarding Valid Sample Comparisons for Erroneous Results

The results do not need to be from the same analyzer or method, unless specifically stated in [MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results](#).

A valid comparison of results consists of any original and repeat test results performed on the same sample from the same tube or multiple tubes drawn at the same time.

- For Near Patient Care (NPC) meters and other devices where the same sample cannot be used to perform repeat testing, this does not apply.
- If the customer alleges the difference in results between two samples is not possible due to the patient's condition, PRI escalation is required.
- If it is not known if the repeat test was from the same sample or new collection, assessment for PRI should be performed.

Results Comparisons that are not subject to PRI escalation for Erroneous Results

- Correlation/Validation studies (e.g., lot-to-lot comparison) are not escalated if the results are not used for diagnostic purposes. If unknown, evaluate for PRI.
- Sample result discrepancies during installation of instruments and until the instrument is fully validated are not evaluated for PRI.
- Spiked or pooled samples (unless included in the product method sheets).
- Roche Controls or other manufactured quality control samples (e.g., Bio-Rad).
- Laboratory self-prepared controls.
- Calibration verification studies and linearity studies, as these specific studies do not simulate patient testing.
- Veterinary samples.
- Results generated when the customer is using expired reagents.
- All proficiency samples are excluded from PRI evaluation for erroneous results (e.g., the external quality control samples from CAP, EQA, NRL, EEQ, UKNEQAS).
- Discrepant results outside of the measuring range but which are still valid, where the comparison value is not outside of the measuring range; no need for PRI ER escalation if the clinical interpretation would be the same.

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Handling Flagged and Unbelievable Results

Results which are unbelievable (e.g., physiologically impossible)

Quantitative results not compatible with human life and are therefore unbelievable are to be evaluated for PRI escalation, because these wrong results may be reportable to the National Competent Authority (NCA). Exception is the list below where the parameters with associated unbelievable results are not to be evaluated for PRI, because they are deemed not reportable to NCA:

- A result of Zero (0) for the following Blood Gas Tests: pH, H⁺, PaO₂, PaCO₂, HCO₃⁻, SBCe, tCO₂, O₂ Content (ctO₂), O₂ Sat (SO₂).
- A result of Zero (0) ISE tests: Na (Sodium), K (Potassium), Ca (Calcium), Cl (Chloride)
- A result of Zero (0) for complete blood count tests: RBC, WBC, Hb, tHb, Hct, platelets, MC, MCHC, MCV.
- A result of Zero (0) for Substrate tests: Calcium
- Negative results, i.e., results below zero, such as -1 or lower, are not believable for all tests, including Elecsys tests, and do not require PRI evaluation.

Roche devices, or instruments, have different behaviors when handling flagged results. Please refer to the guidance below when determining if PRI evaluation is needed for flagged results.

NOTE: Regardless of the presence or absence of a flag, if an erroneous result may have led to an adverse event, the issue must be escalated as a serious injury.

Qualitative results, such as those used in the Pathology Lab, may also be considered unbelievable. Believable results are those that have been released to the patient medical record and/or medical personnel treating the patient and are therefore available for viewing outside of the reporting laboratory, thus are, by definition, believable to the pathologist interpreting the stain for that patient sample.

Immunology Tests

- If the instrument has a problem and cannot give a valid result, the instrument will not give a result. The device will alert the user to the problem by generating a flag that explains WHY the result has been blocked. PRI evaluation is not required in this situation as no results are provided.
- If the device gives a result to the user, the result is considered acceptable, even if it is flagged, and is released by the instrument as a valid result. Evaluation for PRI is required for immunology tests, even if the results are flagged.

Clinical Chemistry, ISE, Blood Gas, Hematology, cobas t 511/711, U701

Erroneous results may contain a flag. Some flags invalidate the result, and some do not.

- Erroneous results that contain a flag, which does NOT invalidate the results, DO require evaluation against the screening list criteria for possible PRI
- Erroneous results with flags that should have invalidated the result do not require evaluation against the screening list criteria for possible PRI.

Refer to the following charts to identify flags that invalidate results:

Clinical Chemistry that invalidates results							
>Abs	ADC.E	Calc.?	CLcT.E	CmpT.?	Det.S	Edited	Hook
>Index	>I.xxx	ISE Unstable	>Kin, >Kin1, >Kin2, >Kin3	>Lin	Mix.E	>Mix	< MIX
MIXLOW	MIXSTP	Over.E	>Proz	>React	>reac0	>reac1	>reac2
ReagEx	Samp.?	SampC	Samp0	Samp.S	<Test - See exceptions below	>Test	< Test Rng
> Test Rng							

Exceptions for Clinical Chemistry "<Test" flag: The following tests must be evaluated regardless of the presence of a < Test flag, as the expected results for these assays may be at or below the test range for that assay:

- Therapeutic Drug Monitoring tests (TDM): See [MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results](#) for TDM tests
- Drugs of Abuse Tests (DAT): See [MQMS-PM-GSP-05-CC-02 - PRI Screening Criteria for Erroneous Results](#) for DAT tests
- CRP/CRPHS
- ASO/ASLO
- Alpha 1-Microglobulin
- Alpha 2-Macroglobulin
- Microalbumin (mALB)
- Rheumatoid Factors (RF)
- D-Dimer

ISE				
Cal.I	>/< ISE	ISE.E	ISE.N	na.LHI
Reag.S.	<>Test	> Test	< Test	< Test Rng

ISE				
> Test Rng				

Blood Gas				
... check Hct result	...improper sample	* ... demo mode	Sensor older than 28 days	...check plausibility
↓↓X	↑↑X	Out of range (-) / (+)	Interferences (1/2/3/4/5/6)	↓↓↓↓ ↑↑↑↑
↓↓↓↓ ↑↑↑↑	I ...			

cobas t511 / t711				
<dABS	>dABS	>Cal	<Cal	>Ext
<Ext	>ABS	>React	Samp.C	Calc.?
>Kin	>Kin1	>Kin2	>Kin3	>Lin
Samp.?	Clot.E	Prob.T	Inc.T	Clot.?
<Test	>Test	No Clot	>CTVal	<CTVal
EXM.T	negABS	CDC.O		

cobas 6500 (u601/u701)				
F1	F2	F3	F4	Ub
Uc	A	Cm	Cp	Cs
H	L	N	O	Q
R				

Hematology				
Damaged cells	Hemolysis?	RBC agglutination?	Too few isolated PLTs for analysis	PLT clumps?
Too few isolated RBCs for analysis	Too few isolated reticulocytes for analysis	RBC discrepancy?	PLT linearity	RBC linearity
HCT linearity	HGB linearity	Retic linearity	WBC linearity	

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Guidance on Calculation of Erroneous Results

The following calculations are examples on how to do the calculation in order to have a common procedural method.

On rare occasions, there may be a combination of results, which may not be mentioned in the following examples. In these cases, the criterion with the lowest number, should be used (worst-case scenario).

General:

- Difference (Δ) = first result - second result (use absolute value)
- (Difference (Δ)/smallest denominator) x 100 = percent difference

If the calculated percentage meets the criterion from the screening list, a PRI is required. For the calculation, the absolute value of the difference is used, independent of being positive or negative.

Scenario	Guidance
A patient sample is tested for Sodium and gives results of 134, 138, 146, and 148.	PRI criteria for Sodium: Error > 5 mmol/L at values < 135 mmol/L, or Error > 5 mmol/L at values > 145 mmol/L Since it is not known which result is correct, the worst case difference would be the difference between the results of 148 and 134. This case would meet PRI criteria based upon the worst-case scenario, because the difference between 148 and 134 is 14. → Criterion that applies is: Error > 5 mmol/L at values > 145 mmol/L

Difference is defined as percent

Bone	Escalation as PRI required when
PTH (1-84)	Error > 30%

Examples:

Initial result 60 pg/mL, repeat result 30 pg/mL: $\Delta = 30$ pg /mL;

• $30/30 \times 100 = 100\%$ **PRI !**

Initial result 40 pg/mL, repeat result 30 pg /mL: $\Delta = 10$ pg /mL;

• $10/30 \times 100 = 33\%$ **PRI !**

Initial result 20 pg /mL, repeat result 30 pg /mL: $\Delta = 10$ pg /mL;

• $10/20 \times 100 = 50\%$ **PRI !**

Difference is defined as percentage in a defined range of values

Substrates	Escalation as PRI required when
Cholesterol	Error > 50% at values 150-300 mg/dL

Examples:

Both results are in the PRI criteria range

- Initial result 180 mg/dL, repeat result 250 mg/dL: $\Delta = 70$ mg/dL;

• $70/180 \times 100 = 39\%$ NOT a PRI

- Initial result 160 mg/dL, repeat result 250 mg/dL: $\Delta = 90$ mg/dL;

• $90/160 \times 100 = 56\%$ **PRI !**

One result is outside and one is inside the range

- Initial result 160 mg/dL, repeat result 315 mg/dL: $\Delta = 155$ mg/dL

• $155/160 \times 100 = 97\%$ **PRI !**

- Initial result 120 mg/dL, repeat result 200 mg/dL: $\Delta = 80$ mg/dL

• $80/120 \times 100 = 67\%$ **PRI !**

One result is outside the lower and one is outside the upper end of the PRI criteria range

- Initial result 120 mg/dL, repeat result 320 mg/dL: $\Delta = 200$ mg/dL

• $200/120 \times 100 = 167\%$ **PRI !**

Both values are outside the upper or lower end of the PRI criteria range

- Initial result 110 mg/dL, repeat result 145 mg/dL; no calculation for PRI required

- Initial result 310 mg/dL, repeat result 415 mg/dL; no calculation for PRI required

Difference is defined as percentage above or below a defined value

Enzymes	Escalation as PRI required when
Alkaline Phosphatase	Error > 50% at values > 100 U/L
Cholinesterase	Error > 30% at values < 2000 U/L

Example: Alkaline Phosphatase

Both results are in the PRI criteria range

- Initial result 120 U/L, repeat result 220 U/L: $\Delta = 100$ U/L;
 - $100 / 120 \times 100 = 83\%$ **PRI!**

Only one result is inside the range

- Initial result 80 U/L, repeat result 175 U/L: $\Delta = 95$ U/L
 - $95 / 80 \times 100 = 119\%$ **PRI !**

No PRI is required when both values are outside the lower end of the PRI criteria range

- Initial result 45 U/L, repeat result 95 U/L; no calculation for PRI required; no PRI

Example: Cholinesterase

No PRI is required when both values are outside the upper end of the PRI criteria range

- Initial result 2100 U/L, repeat result 3000 U/L; no calculation for PRI required; no PRI

Difference is defined as an absolute value and percentage

Substrates	Escalation as PRI required when
Glucose	Error > 15 mg/dL (0.8 mmol/L at values ≤ 50 mg/dL (2.8 mmol/L)) Error > 30% at values > 50 mg/L (2.8 mmol/L)

Besides the calculations explained above, one special case has to be considered:

Example: Results in the range of the “absolute” and “percentage” value; in this case the absolute value should be taken for PRI calculation

- Initial result 40 mg/dL, repeat result 60mg/dL: $\Delta = 20$ mg/dL
 - Error > 15 mg/dL **PRI !**
- Initial result 70 mg/dL, repeat result 40 mg/dL: $\Delta = 30$ mg/dL
 - Error > 15 mg/dL **PRI !**

Interpretation of further characteristics: PRI criteria may be based on gender or age of the patient e.g., for CK-MB (men or women) or Theophylline (babies up to 6 months).

Process if the gender is not known: Use the worst-case scenario, which in this case would be the criterion for women.

Cardiac Markers	Escalation as PRI required when
CK-MB (mass)	Women: Error > 4.9 ng/mL Men: Error > 8.8 ng/mL

Process if the age of the patient is not known: Use the criterion for individuals older than 6 months because the vast majority of samples are from patients older than 6 months.*

Therapeutic Drug Monitoring	Escalation as PRI required when
Theophylline	Error > 20% at values > 20 µg/mL (111 µmol/L) Error > 20% at values < 6 µg/mL (33.3 µmol/L) Babies (up to 6 months including prematures): Error > 20% at values > 6 µg/mL (33.3 µmol/L)

*Using Gaussian distribution to look at a normal distribution of values, the babies under 6 months would be at the outermost edge of the curve. Thus, it is mostly likely that the age of the individual is older than 6 months.

Decimal Placement

When calculating the difference between the two values, it should be done according to the decimal places provided by the customer. The number should be rounded according to the decimal placement in the PRI criterion.

Example: For Albumin, that means rounding to one decimal placement because the criterion is 0.5 g/dL; e.g., 0.69 g/dL rounds up to 0.7 g/dL.

If the calculated difference has more decimal placements than the criteria, the difference should be rounded using the number of decimal placements provided.

Substrates	Escalation as PRI required when
Albumin (serum, plasma)	Error > 0.5 g/dL at values < 3 g/dL Error > 30% at values > 5 g/dL

Example

- Initial result 1.45 g/dL, repeat result 0.76 g/dL
 - $\Delta = 0.69$ g/dL; rounded = 0.7 g/dL **PRI !**

Criterion 0.5 g/dL: Round up to a number with one decimal place according to the given criterion, e.g., 0.7 g/dL

Criterion 30%: Round up to a whole number e.g., 35%

Malfunction

The following Table describes examples of situations that may require escalation as a PRI for malfunctions along with additional guidance to support the decision-making process.

Types of Malfunction Events	Examples	Additional Guidance
A Roche Diagnostics product created a hazardous environment.	Smoke, noxious odors, or chemical vapors in sufficient quantity that require evacuation.	• If there was no evacuation but evidence of electrical and chemical smells do not escalate as PRI unless the user(s) is harmed or requires medical attention (see 'serious injury or serious deterioration of health' examples for additional guidance). • Leaking or damaged/swollen batteries in NPC meters do not require escalation as PRI. • Uninterrupted Power Supply (UPS) units, monitors, printers, external power supplies do not require escalation as PRI. • Sample spills do not require escalation as PRI because in general the laboratory is a hazardous environment. In the event a sample or reagent spill is actually caused by a Roche device AND the customer requires professional medical treatment (excluding washing of affected area) then a PRI is required.
Evidence of fire on Roche device.	• Fire caused by a Roche device. • Burnt, blackened, melted or scorched instrument panels, or internal components. • Sparks that occur on external electronic components require escalation as PRI.	• Burnt/melted fuses/fuse holders do not require escalation as PRI as this is expected behavior for a blown fuse. • In some situations, heating elements are a normal part of the device. Melting, in these situations, is not a PRI. For examples, melted rims of tubes, PCR plates, or other plastic components.

Types of Malfunction Events	Examples	Additional Guidance
Product labeling (e.g., instructions for use, package labeling, service manual) issues that prevent proper product identification and may impact patient results, interpretation of patient results or product(s) is missing hazardous labeling warning(s). Counterfeit/Modified.	• Expiration date or lot number is missing. • Expiration date or Lot number is inconsistent (e.g., outer packaging indicates a particular lot for R1 bottle but R1 bottle has a different lot number on it). • Incorrect conversion factors. • Labeling incorrectly describes how to prepare a sample for testing or perform the test (i.e. a step is missing or incorrect). • Barcode issue where the wrong product information is scanned without any flag or alarm to alert the user of the problem. • Assay Tip/Assay Cup tray product label is missing. Service manual does not provide clear information to properly configure the device or product. • Operators Manual provides incorrect instructions for device operation or maintenance which could harm the operator of the device.	For SD Biosensor Tests Only: If any of the following circumstances apply, a PRI is not required: • If the instructions for use or quick reference guide are missing. • If the expiration date / lot number label on the outer KIT box is missing. • If the product is not usable due to a missing or non-functioning barcode label (i.e. the instrument is not able to recognize the reagent due to a missing barcode or a barcoded reagent cannot be placed on the instrument and used for testing), and no other labeling issue, escalation as PRI is not required as the missing or non-functioning barcode prevents the use of the affected reagent, thus the missing label does not compromise end-user safety or patient care. • If the label comes off the product or it is difficult to read because of a handling issue (e.g., cleaning with wrong disinfectant) a PRI is not required. • Typographical errors, spelling errors, and translation errors only require escalation as PRI if the incorrect information prevents proper product identification or may impact patient results or interpretation of patient results (e.g., the translation incorrectly describes sample preparation or dilution).

Types of Malfunction Events	Examples	Additional Guidance
Packaging Issues. Sterility.	Packaging issues do not require classification as PRI unless there is harm to the patient (as defined in Serious Injury section) or erroneous results occur (as defined in erroneous results section).	Logistics claims should not be classified as PRIs. If there is no information regarding patient harm, a PRI is not required.
Software issues that lead to incorrect assignment of patients' surnames, patient Date of Birth (DOB), Age, or Gender.	• Wrong patient ID, name, surname may cause results to be assigned to another patient. • Wrong birth date, age, gender affect the correct validation of results. • Editing a patient's demographics changes another patient's details. • When merging patients, incorrect demographics are assigned to the target patient.	Typographical errors by the user are not device malfunctions and do not require escalation as PRI.
Software issues related to transformation algorithms impacting results.	• The transformation is triggered when it should not, causing a result to be modified. • Modifying a transformation rule also changes another rule and the user is not aware. • The transformation rule is adding a comment/flag to an incorrect test.	N/A

Types of Malfunction Events	Examples	Additional Guidance
Instrument or software problems which cause an incorrect protocol/parameters to be used when performing/handling tests.	• Missing/wrong command or step that could cause cross contamination. • Wrong staining protocol applied on tissue sample.	N/A
Software issues that lead to wrong or missing information available for patients or Health Care Providers (HCPs) that could lead to misinterpretation of health status or actions to be taken.	• Comments in blood depot module are lost or truncated impacting correct selection of donor blood. • Lost or truncated comments that contain information on flags and/or alarms that are needed for interpretation of associated results. • Comments field contains results. • Report includes incorrect patient demographics (e.g., the ones corresponding to a previous order). • Incorrect reference ranges shown.	For slide staining, the following situations do not require escalation as PRI: • If the information is changed within the software after a slide label is printed, a PRI is not required even though the results and software generated reports may not match the initial label. The system will run the tests according to the most updated information in the software. • The master computer where labels are printed has an incorrect date.
Problems with software alarms or flags potentially causing a misinterpretation of the results.	• Flags/alarms are incorrect or assigned to the wrong test result. • Flags/alarms are missing. • The result is correct but the alarm/flag is incorrect.	If customer states that an alarm or flag is missing and through troubleshooting it is determined that the device is working as intended, this is not a PRI.

Types of Malfunction Events	Examples	Additional Guidance
Consumables with product defects that impact patient results.	- Incubation cups with holes. - Closed assay tips.	N/A
Unplanned/extended instrument downtime and/or reagent unavailability impacting routine/emergency operation, with impact on the patient treatment.	Delayed results due to unplanned, extended instrument downtime or reagent unavailability impacting routine/emergency operation, leading to a delay in required medical treatment.	If not known if there is an impact to a patient, document but do not escalate as PRI.

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Near Patient Care-Specific Malfunctions

The following table describes examples of malfunctions that may require escalation as a PRI along with additional guidance to support the decision-making process

Types of Events	Examples	Additional Guidance
Meter Display issues that interfere with the user's ability to correctly read test results on the display AND the customer alleges that a misinterpretation of results with an impact on a patient has actually occurred.	The customer alleges a misinterpreted result that had caused harm to the patient (e.g., changed/delayed medication/treatment) due to a display issue, e.g., - The meter display is missing segments so that it is unclear what the number(s) should be (e.g., an 8 is missing segments and looks like a 3). - The meter display has lines through the results section so that it is unclear what the number(s) should be (e.g., there is a line in the results section where a 1 looks like a 7). - Faded display if it is unclear what the number(s) should be.	- Display issues without allegation of an impact to a patient (i.e. the user recognized the display issue but did not act upon it) no longer qualify as PRI. - Consider classification as PSI depending on the individual case (see guidance above). - If the customer or patient does not know if there was an impact to the patient, then the case does not qualify as PRI.

Types of Events	Examples	Additional Guidance
Properly seated lancet does not fully retract and extends beyond a correctly positioned end cap of the lancing device.	N/A	N/A
An uncovered lancet needle falls out of a container drum or is exposed.	N/A	N/A
Lancets from a newly opened box are uncapped or exposed.	N/A	N/A
Lancet or lancet fragment remains in the patient or individual after skin puncture to obtain a blood sample.	N/A	N/A
Individual is accidentally stuck with a previously used lancet.	N/A	Refer to Serious Injury / Serious Deterioration of Health Section - Direct Injuries - Accidental Needlestick to determine if PSI is needed.
Failure of device control buttons or touch screen.	A button(s) or touch screen temporarily or permanently fail(s) or was/were difficult to press, slow to respond or sluggish on an insulin pump (e.g., Accu-Chek Insight or Spirit Combo) or the remote control for the Accu-Chek Solo micropump.	Troubleshooting resolves the button concern, no defect or device failure occurred. The temporary or permanent failure of any button function after report drop or immersion in water (that means the pump was completely covered with water for example in the pool, during a bath in the bathtub, while taking a shower etc.). ACCU-CHEK Spirit Combo: complaints related to a visually damaged, or missing, soft component. ACCU-CHEK Solo Guide Diabetes Manager: Any problem with the EJECT button, Meter hang up / freeze. All non-Roche devices are excluded.

Product-Specific Malfunctions Requiring Escalation as PRI

This section includes unique situations where specific criteria must be applied in relation to a confirmed product issue resulting in field action.

BenchMark Ultra and DISCOVERY Ultra devices require a PRI for leaking ONLY if BOTH of the following occur:

- The BenchMark Ultra or DISCOVERY Ultra waste tub overflow sensors fail to indicate a waste tub overflow condition (error message 12-19 or 12-20 Fluid in tub exceeded maximum level).

AND

- Fluid leaks from the waste tub to the floor.

Cybersecurity or Data Breach

Cybersecurity (CS): The protection of a device and its data from unauthorized access, control or manipulation

Data Breach: An event leading to the accidental or unlawful destruction, loss, alteration, unauthorized disclosure of, or access to, personal data transmitted, stored or otherwise processed. A data breach is alleged to have been caused by (or via) a CS. In either case, a Data Breach should be reported by the local organization to the country affiliate's local privacy officer.

When the customer mentions a possible cybersecurity or data breach issue, ask for information that will help determine if the issue meets PRI criteria and will support further investigation as needed:

- Were there any harm or health consequences to patients?
- Was user data extracted from the device, breached by third parties, or released to third parties?
- Was the device remotely operated, influenced, or altered by unauthorized parties?
- Was data communication altered by unauthorized parties?

Types of Events	Examples	Additional Guidance
CS issues or other data breaches where the Roche device is the cause or has been directly impacted AND potential harm to patients and/or users could occur.	• Patient received an incorrect diagnosis due to missing or incorrect information. • Results are assigned to incorrect patient. • Unauthorized sharing of patient identifiable data and/or patient results due to a CS or data breach issue. • Patient Identification and/or Demographics are assigned to the incorrect patient. • Device is inoperable due to a CS issue. • Unauthorized remote operation of a Roche device.	• CS issues or other data breaches where the Roche device is the cause or has been directly impacted and no potential harm to patients and/or users could occur, issue must be escalated as a complaint but not PRI. • If the device is not available for use because the customer took the device offline as a precautionary measure, this is not a PRI. Refer to Serious Injury Serious Deterioration of Health Section, Delays, if device is unavailable for use due to a CS issue.

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Notification of the Competent Authority

The criterion **does not** apply to the following countries:

- Australia
- China

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Document History

Document history on the effective document will not exceed the last 5 versions. Refer to older versions for a comprehensive history of the content.

Version Number	Date	Author	Description of Changes	Justification of Changes
5	19 Dec 2023	Elswe dy, Aman y {DQM E~BR ANCH BURG}	- Added 'Note' regarding RUO, LUO, Analyte Specific Reagents, Industrial Business or any additional non-In Vitro Diagnostic test commercially distributed by Roche Diagnostics. - Added CalCt.E and >Kin1, >Kin2, >Kin3 flags to the Clinical Chemistry table and rearranged flags in table. - Directed users to MQMS-PM-GSP-05-CC-02 to find Therapeutic Drug Monitoring tests (TDM) and Drugs of Abuse Tests (DAT). - Added 'I ...' flag to the Blood Gas table. - Corrected the product (calculation) to an example - Deleted duplicate calculation - Updated 'Additional Guidance' for the 'Malfunction Event' example: A RD product created a hazardous environment. - Updated 'Meter Display issue' example. (REQ00260C)	- Guidance on RUO, LUO, Analyte Specific Reagents, Industrial Business or any additional non-In Vitro Diagnostic test commercially distributed by Roche Diagnostics was only captured in MQMS-PM-GSP-05-CC-02 for erroneous results; guidance was not provided for other non-erroneous results issue. - R&D confirmed, that the CalCt.E flag is triggered by a <>Test flag which is already in the list. On the analyzer only one flag can be displayed meaning the flags have certain priorities. As the <>Test flag has a lower priority as the CalCt.E, only CalCt.E is displayed although the result has for example both flags. It means that the result cannot be used because it is calculated incorrectly. >Kin1, >Kin2, >Kin3 was added for clarification. Table rearranged to capture flags in alphabet order. - Clarifies which tests fall under TDM and DAT. - Aligns with the cobas b 221 system Operators Manual. - The product had to be rounded up due to the answer. - Calculation is captured under appropriate example. - Update is required for clarification on when a case must be escalated as PRI. - Update is required for clarification.

Version Number	Date	Author	Description of Changes	Justification of Changes
6	04 Oct 2024	Elswe dy, Aman y {DQM E~BR ANCH BURG}	1.Moved and added additional information regarding a serious injury issue. 2.Clarified the prophylaxis treatment example. 3.Separated the bullet capturing 'were erroneous results reported to the patient and/or medical personnel treating the patient' into two different bullets and added a note regarding medical record definition 4.Moved 'Note' regarding when the Erroneous Results examples do not apply 5.Deleted 'testing is not repeated' under the Erroneous Result scenario 6.Added additional scenarios under Erroneous Results 7.Added additional guidance (including which products are excluded) for allegations of erroneous results in standalone software products 8.Deleted table under 'Information Regarding Valid Sample Comparisons for Erroneous Results' 9.Updated guidance regarding Correlation/ Validation studies 10.Added information regarding Emergency Use Authorization (EUA) products. 11.Deleted last bullet under 'Results Comparisons that are not subject to PRI Evaluation for Erroneous Results' 12.Added information regarding qualitative results 13.Corrected Clinical	1.Cases must be marked a PSI only if there is confirmation of harm/impact on the patient is made. The clarification helps users avoid escalating cases where there is no 'real' allegation of harm. Moved information for a better flow in document. 2. To capture potential PHT cases due to infectious diseases 3.Information regarding 'medical record' is captured throughout the procedure. The note clarifies what a medical record entails. Bullet was separated into two to capture each question into a separate bullet. 4.Flows better within the document 5.Repeat testing does not support the decision-making process when marking cases a PRI 6.Provides users with additional scenarios that are not currently captured. 7. To avoid over-classifying cases as PRI when it's not required. 8.Information already captured in the text above. 9.Clarified when Correlation/Validation studies are not subject to PRI evaluation. Current wording was confusing. 10.EUA products must be evaluated for PRI 11.Scenario is already described in the section "Guidance on Calculation of Erroneous Results". 12.'Handling Flagged and Unbelievable Results' currently only captures additional information regarding Quantitative results 13.Captured what users see 14.The calculations were incorrect 15.Information adds additional guidance to users. This will avoid over-escalated cases as PRIs and clarifies what is meant by 'impact to the patient' 16.Aligned with wording seen under 'Types of Events' 17.Improve document (REQ00588C)

Version Number	Date	Author	Description of Changes	Justification of Changes
			Chemistry Flag 14. Corrected calculations 15. Added examples to meter display examples, added additional guidance, and clarified wording 16. Reworded additional guidance for cybersecurity issue 17. Fixed typos and grammar issues (REQ00588C)	

Version Number	Date	Author	Description of Changes	Justification of Changes
7		Elswe dy, Aman y {DQM E~BR ANCH BURG}	1. Removed/updated tool-related terms/content, including title, 'Purpose', PRI/PSI terminology and removing the category questions. 2. Added a 'NOTE' under 'Purpose' and removed EUA info under Result Comparison section. 3. Renamed 'Point of Care' to 'Near Patient Care' 4. Added 'Notification of the Competent Authority' category and added 'Counterfeit/Modified' as type of malfunction event. 5. Merged examples falling under 'Types of Serious Events' 6. Additional examples and types added for integrated entities. 7. Removed mQMS title names. 8. Clarification throughout document, including updating/deleting acronyms and formatting updates were made. 9. Deleted 'NOTE' under Purpose. (REQ00778R)	Update for Fusion integration and continuous improvement: 1. Other Case Handling IT systems will be used by integrated entities. 2. To capture that the FDA can add additional requirements for EUA products that may require escalation as a PRI. 3. To reflect the name change of the organization. 4. To align with MQMS-PM-GSP-05. 5. Examples are related to one another. 6. To capture guidance for users 7. Document number is captured. 8. To enhance clarity and improve the overall flow. 9. This version had several updates due to integration and continuous improvement. (REQ00778R)
8	03 Apr 2025	Elswe dy, Aman y {DQM E~BR ANCH BURG}	Correct the "Description" metadata field to match current document's description/title. (REQ00819R)	Ensure metadata is correct and matches the document. Translations are not impacted. (REQ00819R)

Version Number	Date	Author	Description of Changes	Justification of Changes
9	see eDMS	@Aman y Elswedy	1. Changed title from all caps to sentence case. 2. Added navigation markers throughout document. 3. Aligned PRI/PSI terminology throughout, including in title of document. 4. "Serious Injury / Serious Deterioration of Health" changed to "Serious Injury / Serious Deterioration of the Status of Health". 5. New clarifying questions added to "Death" category. 6. Additional guidance sections updated in "Serious Events", "Erroneous Results", and examples updated in text section after "Erroneous Results" tables. 7. Added note for self-administered treatment in "Serious Events" table. 8. Updated Notes before guidance table for Erroneous Results. 9. Added "Hook" in Clinical Chemistry that invalidates results table. 10. Calculation of sodium values example moved from Erroneous results table to Guidance on Calculation of Erroneous Results section.	1. Alignment with other documents' style. 2. Enhancement for ease of use. 3. Alignment across and within documents. 4. Clarification of intent. 5. Enhancement. 6. Clarification and alignment, to reduce incorrect / unnecessary escalation of PRIs. 7. Alignment with UK legislation requirements for blood glucose testing. 8. To align with changes made in MQMS-PM-GSP-05-CC-02 table headers. 9. Additional parameter. 10. Clarification. 11. Improvements. (REQ01311R)

Version Number	Date	Author	Description of Changes	Justification of Changes
			11. Typographical and grammar changes throughout document. (REQ01311R)	